Labs 4: Geiger Muller Counting

10/15/2026

Reading Notes

GM Tube Physics (Ch7 pp. 207-212, Ch 6 pp. 159-163

Primary Ionization: · charged particle enters gas, creates ion pairs along its track (Coulomb collisions)

$E(r) = \frac{V}{r \ln(b/a)}$ (Eq. 6.3)

· one pair enough to discharge (chain reaction, not detector efficiency of e^- gain energy & ionize more gas) (efficiency is 100%, every particle that enters is counted)

→ charged particle hits to reach gas (get through window w/o abs or backscatter)

· Field grows a 1/r toward wire, gas discharge has E field threshold
· b inner cathode radius
· a outer anode radius
· decrease a, put multiplying region in thin shell close to wire

· γ ray efficiency few % b/c only counted if interact w/ wall & secondary e^- escapes into gas
· still affected by dead time, efficiency ≠ dead time

· Townsend Avalanche: e^- in motion ionizes other atoms

$\frac{dn}{n} = \alpha dx \Rightarrow n(x) = n(0)e^{\alpha x}$ Eq. 6.1, 6.2

1st TS coefficient $\propto E$ energy E field

n: # excited molecules

$n''p \ll 1$ proportional, subcritical

$n''p \gg 1$ Geiger: critical, chain reaction

$\alpha \propto E$

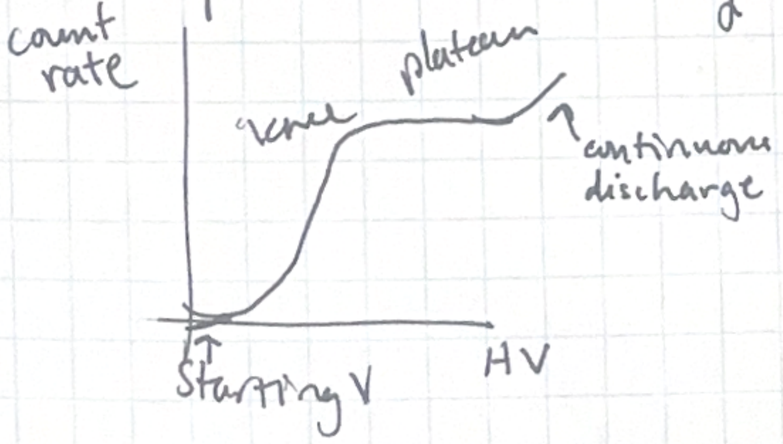
· GM tubes sealed @ lower pressure, $E_{c,threshold} \propto E/p$
· Avalanche exites molecules, emit photons $\Delta t \sim$ ns, 10^{-9}
→ photon photoionizes gas, or release e^-, new avalanche
· discharge speeds 2.4 cm/μs, all charge made ~μs
· $+i$ slower, form space charge sheath that acts like larger effective wire radius
· required n fixed by V, GM tube is counter

$E = |E - W$ Quenching · ions → cathode neutralized & release energy
· if $|E > 2W(f)$, e^- emitted
external quenching: drop V below multiplication threshold, full discharge occur @ low rates
internal " : 5-10% of another gas

Counting Curve: gain v. counts @ fixed Hd

Prelabs Q's

1) Shape of GM curve & why:



· @ starting V, pulses @ Hd begin to discr.
· rises steeply as ~~go~~ across DPHS
· knee: transition into plateau, every source exceeds Hd
· @ Plateau, rate is flate, raising V just moves DPHS peak →
· never perfect flat b/c low amplitude tail
· @ end is continuous discharge

2) what sets pulse height:

· applied V, tube geometry (a, b), fill gas, pressure, quench gas
· external circuit ($\tau = RC$)

3) Error of net count rate:

$r = \frac{G}{t_G} - \frac{B}{t_B}$ $\sigma_r = \sqrt{\frac{G}{t_G^2} + \frac{B}{t_B^2}}$

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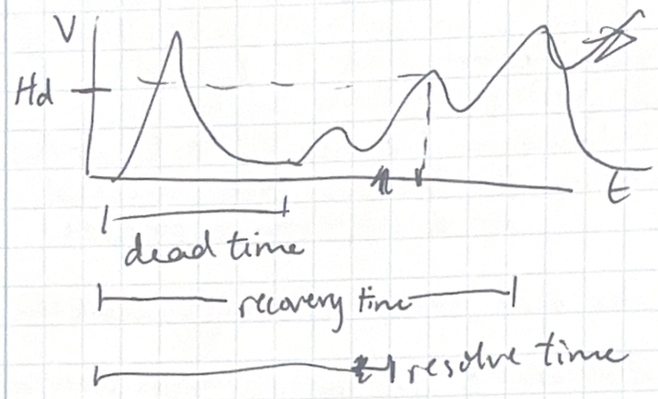
4) dead time, ~~resolving~~ resolving time, ~~recovery~~ recovery time.
& dead time τ: t until GM discharge can develop another
resolving ~~recovery~~ time: until 2nd pulse large enough to discr.
recovery time: until field fully restored & 2nd pulse full amp



$t_{dead} < t_{resolve} < t_{recovery}$

5) Paralyzable v. non-paralyzable: n: true rate m: recorded rate
τ: dead time

paralyzable: $m = ne^{-n\tau}$ — τ extended w/ every count

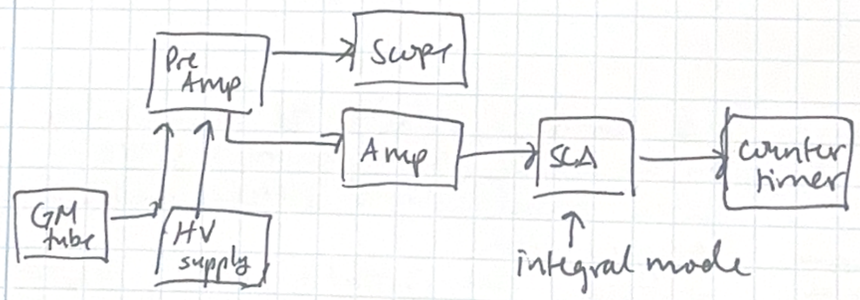
non-paralyzable: $n = \frac{m}{1-m\tau}$ — fixed τ

@ low rates: both models agree
@ high rates: nonpara saturates

Partner: Charalampos, 10/15/2026

2) Counting Curve

Equipment: GM tube, thorium source, oscilloscope, pre-amplifier, amplifier SCA analyzer, counter/timer, MCA



Procedure: Following Lab Manual 4 (Fall 2026)

2.2.1 Determine LLD

- Set HV = 800V
- remove source
- Find LLD high enough to discriminate noise

$LLD_{noise} = 0.1V$

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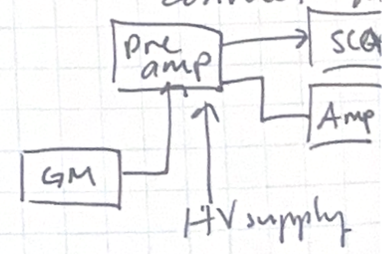
2) 2.2.2 Generat
- put source
- reduce HV
- inc. HV in
Need: co
- repeat 10

count inte
scope
Start
at beginning: Record
determi
for avg: fu

HV (V)	650	670	69
counts	0	44	1
Pulse Height (mV)	0	0	18.7

HV (V)	790	810	8
counts	169	178	1
Pulse Height (mV)	137.5 ± 28.75	155.31 ± 29.0	21 ±

3) Pulse Heig
- Select id
Choos
- connect



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2) 2.2.2 Generate counting curve

- put source in chamber SOURCE used: thorium-232 ↳ height ~5cm
- reduce HV until almost no pulses
- inc. HV in increments of 10-30 V
  Need: count rate, avg pulse height from pre-amp
- repeat 10 times, wait ~30 before recording

count interval: $\Delta t_c = 3$ sec

scope baseline: $V_{base} = -4$ ~~00 mV ± 2.625~~ -2 ± 2.625 mV (10/15 CL)

Starting HV: 650V

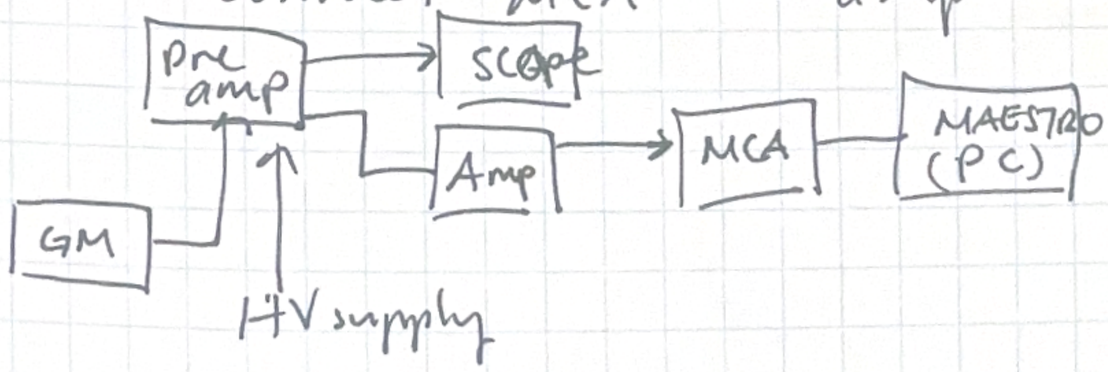
at beginning: Record ~~&~~ baseline for ~~amplitude~~ avg pulse height
determine count interval for count rate
for avg: turn on scope persist

HV (V)	650	670	690	710	730	750	770
counts	0	44	119	152	163	164	169 ~~190~~
Pulse Height (mV)	0	0	18.75 ± 2.75	31.125 ± 4.13	47.88 ± 9.13	70.5 ± 10.75	97.63 ~~75~~ ± 14.38

HV (V)	790	810	830	850	870	890	910
counts	169	178	183	185	178	177	267
Pulse Height (mV)	137.5 ± 28.75	155.31 ± 29.06	212.81 ± 19.06	255.0 ± 21.25	303.75 ± 35.0	343.75 ± 52.5	443.75 ± 56.25

3) Pulse Height consistency

- select ideal operating V from curve
  (choose middle of plateau) V_{op} = ~~760 V~~ 840V (10/12 CL)
- connect MCA to amplifier



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3) For 3 sources, record 30s spectrum
measurements: pulse amplitude centroid, gross counts

	Co-60	C-14	Cl-Bb
10/12 CL	~~Th-232 Thorium~~		
~~per~~ channel # ~~amp~~: ~~3374~~ 10/12	2906	3374	1939
X cts:	4419	1280	13986
$\sqrt{x}$ uncertainty:		$\sqrt{1280} = 35.78$	
Convert channel # n:	3.83V	4.44 V	2.50 V

to V w/ ~~V=~~ ~~2755mV + (76.5.76mV) n~~ 10/12 CL

$V(n) = 35.98\,mV + (1.30\,mV)\,n$

4) Beta Attenuation, estimate absorption coefficient $\alpha \left[\frac{abs}{cm}\right]$

- Detector @ ideal operating voltage $V_{op} = 840$ V
- Choose β source w/ ~1K counts/sec on 2nd tray: Co-60 in CL 10/12 Fig 2.4 book
- Background counts: ~~at~~ $T_B = 5s$ ~~B~~ $N_B = 5$

calculate slope, compare w/ book

[al count table]

nxm table, n = 8 m: ~~Ex~~ thickness, counts, etc

Use SCA

5) Dead time & Recovery time: measure dead time 2 ways

5.1) Oscilloscope Dead Time Measurements Co-60

- Source on top shelf
- V_{op}: 840 V

dead time: $\tau = 360\,\mu s$
recovery time: $t_{recovery} = 482\,\mu s$

[screenshot from scope]

print_008 : dead time
print_009: recovery time

C

Cl Br
Ca

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5.2) Two-source Deadtime measurement

V_{op} = 840 V Use SCA

· measure $B_g = \frac{N_{Bg}}{T_{Bg}}$ N_{Bg} = 2 T_{Bg} = ~~5~~ 10 s

10/12 CL

~~SOURCE A~~: Source ~~A~~ #1: Th Source #2: Cl-36

~~A~~

Source 1+2

N_{12} : N_1 : 968 N_2 :

T_{12} : T_1 : 8 s T_2 : 8 s 3rd shelf (22cm) side by side

$S_1 = \frac{N_1}{T_1}$:

$S_2 = \frac{N_2}{T_2}$: repeat 3 times, make spreadsheets

$S_{12} = \frac{N_{12}}{T_{12}}$: [TABLE]

· measure source 1 only, source 2 only, source 1+2, background

→ repeat 3 times